
Osteoarthritis Imaging in the Stone Age: An Investigation Using Prehistoric Scanning Methods

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Abstract

This example thesis demonstrates the automatic features of the University of Brighton template. It examines a hypothetical study of quantitative magnetic resonance imaging for identifying early osteoarthritis. The document contains chapters, sections, equations, tables, figures, captions, cross-references, citations and appendices. LaTeX generates their numbering and the associated contents pages automatically.

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Acronyms and definitions

MRI

Magnetic resonance imaging.

OA

Osteoarthritis.

Preface

Delete this file and its corresponding `\include` line in `main.tex` if a preface is not required.

Acknowledgements

Add acknowledgements here.

Declaration

I declare that the research contained in this thesis, unless otherwise formally indicated within the text, is the original work of the author. The thesis has not been previously submitted to this or any other university for a degree, and does not incorporate any material already submitted for a degree.

Signed: _____

Date: _____

Chapter 1

Introduction

1.1 Background

Osteoarthritis is commonly recognised after symptoms or structural changes become established. A hypothetical research programme might test whether quantitative magnetic resonance imaging can identify earlier tissue changes. The condition affects the whole joint rather than cartilage alone, so a useful study would combine structural, compositional, mechanical and patient-reported measures.

This sentence demonstrates a footnote.¹

The example bibliography is demonstrated using Example Author (2026).

Figure 1.1 demonstrates automatic figure numbering. Its caption also creates an entry in the list of figures.

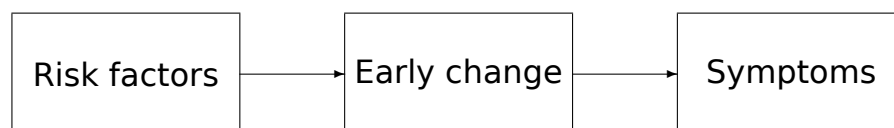


Figure 1.1: Illustrative pathway from risk factors to symptomatic disease

¹Footnotes appear at the bottom of the page in smaller, single-spaced text. They are numbered automatically and should contain information needed to understand the corresponding passage.

1.2 Rationale

Early identification is valuable only if an imaging signal can be reproduced, interpreted clinically and associated with a meaningful later outcome. Consequently, an example thesis might distinguish three linked questions:

1. Can early tissue change be measured reliably?
2. Does that measure predict symptoms or structural progression?
3. Does it add information beyond routine clinical assessment?

1.2.1 Clinical context

Osteoarthritis illustrates why a single observation rarely provides a complete account of disease. Symptoms, structural findings and function may change at different rates. A clinically useful study must therefore explain who is being assessed, when assessment occurs and how an imaging result might alter a decision. The following demonstrates an unordered list:

- symptoms reported by the participant;
- structural features visible on conventional imaging;
- compositional features measured using quantitative MRI; and
- contextual factors such as age, previous injury and mechanical load.

1.2.1.1 Symptoms and function

Pain intensity, stiffness and activity limitation describe related but distinct parts of the participant's experience. A thesis should define each outcome, identify the instrument used and state the time point to which it relates.

1.2.1.2 Structural assessment

Radiographs and morphological MRI sequences can describe established structural features. Their interpretation should include information about acquisition, reader training and repeatability.

1.2.1.3 Compositional assessment

Quantitative sequences may be sensitive to tissue changes that precede gross structural loss. Their value depends on measurement reliability and evidence that the signal predicts an outcome important to patients. This is the third subsubsection under Section 1.2.1, so its automatically generated number ends in “.3”.

1.2.2 Conceptual framework

A conceptual framework makes assumptions visible before analysis. In this example, exposure and susceptibility factors may influence early tissue change, which may in turn affect later symptoms and function. Potential confounders should be identified from subject knowledge rather than selected solely from observed statistical associations.

1.2.3 Translation to study design

The framework can then be translated into a sampling strategy, measurement schedule and analysis plan. This subsection is deliberately included as the third subsection of Section 1.2, demonstrating an automatically generated heading numbered 1.2.3.

Figure 1.2 demonstrates a more detailed diagram created directly in LaTeX using TikZ. Because it is vector artwork, it remains sharp at any zoom level.

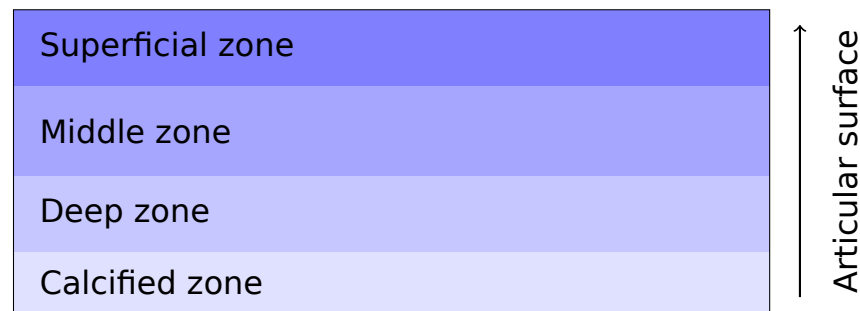


Figure 1.2: Schematic representation of the zones of articular cartilage

1.3 Research question

State the primary research question and objectives. A simple mathematical example is

$$R = \alpha I + \beta M + \gamma S. \quad (1.1)$$

Indented quotations and footnotes may use single spacing under the Brighton requirements.

Equation 1.1 is numbered automatically and can be cited from any chapter.

1.3.1 Example objectives

The description-list format is useful when each item needs a short label followed by an explanation:

Objective 1

Estimate the repeatability of the quantitative imaging measure under a prespecified acquisition protocol.

Objective 2

Test its association with change in symptoms and structure over 24 months.

Objective 3

Assess whether it adds useful predictive information to routinely available clinical variables.

1.4 Chapter summary

The chapter has demonstrated several paragraphs of ordinary prose, three levels of numbered heading, ordered, unordered and description lists, two types of figure, a quotation, a footnote, an equation and cross-references. The source file remains independent from the other chapters while all numbering is controlled by `main.tex`.

Chapter 2

Methods

2.1 Study design

Describe the study population, sampling, imaging procedures, outcomes and analysis plan. Table 2.1 demonstrates an ordinary floating table. LaTeX supplies the chapter-based number and adds the caption to the list of tables.

For illustration, assume a prospective cohort with baseline and 24-month follow-up. Participants would undergo symptom assessment, clinical examination and a standardised imaging protocol. An analysis plan written before outcome assessment would identify the primary measure, expected confounders and approach to missing observations.

2.1.1 Eligibility

An example set of inclusion criteria might require adults with knee symptoms but no established radiographic disease. Exclusion criteria could include contraindications to MRI, inflammatory arthritis, recent major trauma or inability to complete follow-up.

2.1.1.1 Recruitment pathway

Potential participants could be identified through primary care, community advertising or existing research cohorts. Screening should use the same written criteria in every setting, and the number

approached, assessed and enrolled should be retained for the participant-flow diagram.

2.1.1.2 Consent and withdrawal

The information sheet should explain the imaging procedure, use of clinical data, foreseeable burdens and the right to withdraw. The thesis should distinguish withdrawal from further contact, withdrawal from future data collection and any request concerning data already obtained.

2.1.2 Imaging acquisition

Scanner field strength, coils, positioning, sequences, voxel dimensions and quality-control procedures should be reported sufficiently clearly for the method to be reproduced.

Table 2.1: Example research measures

Domain	Measure	Purpose
Imaging	Cartilage thickness	Structural change
Clinical	Pain score	Patient experience
Mechanical	Knee load	Physical exposure

2.2 Multi-page tables

The `longtable` environment repeats its headings automatically when a table crosses a page boundary. The short example below uses the same structure and can be extended with additional rows.

Table 2.2: Example data dictionary

Variable	Type	Description
Participant ID	Text	Pseudonymous study identifier.
Age	Number	Age at baseline assessment.
Pain score	Number	Patient-reported symptom measure.
Cartilage thickness	Number	Quantitative structural imaging measure.
T2 relaxation time	Number	Quantitative compositional imaging measure.

2.3 Analysis

Continuous variables could be summarised using means and standard deviations or medians and interquartile ranges, depending on their distributions. Model assumptions should be checked and uncertainty reported using confidence intervals rather than relying only on thresholded significance tests.

$$Y_{24} = \beta_0 + \beta_1 X_{\text{MRI}} + \beta_2 Y_0 + \beta_3 A + \varepsilon, \quad (2.1)$$

Equation 2.1 illustrates a follow-up outcome adjusted for baseline outcome Y_0 , age A , and an MRI predictor.

2.3.1 Sensitivity analyses

Sensitivity analyses might repeat the primary model after alternative handling of missing data, exclusion of scans that failed quality control and adjustment for additional prespecified confounders. These analyses should test the robustness of interpretation rather than act as a search for a preferred result.

2.3.2 Reporting

Results should be presented with denominators, effect estimates and confidence intervals. Tables should be understandable without repeatedly consulting the main text, while the narrative should highlight the pattern of findings rather than reproduce every number in the table.

Chapter 3

Results

3.1 Participant flow

Figure 3.1 provides a second automatically numbered figure.

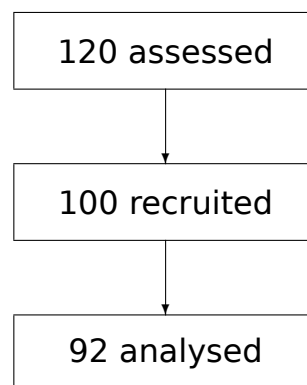


Figure 3.1: Illustrative participant flow through the study

3.2 Illustrative findings

The hypothetical imaging measure changed between baseline and follow-up. Replace this example with the actual analysis and its corresponding tables or figures.

Table 3.1 shows the recommended practice of giving the table number and title above the table. Units are included in the row labels, and missingness is reported explicitly.

Table 3.1: Illustrative baseline characteristics

Characteristic	Analysed sample	Missing
Participants, n	92	–
Age, mean (SD), years	57.4 (8.6)	0
Female, n (%)	54 (58.7)	0
Pain score, mean (SD)	3.8 (1.9)	2
Cartilage thickness, mean (SD), mm	2.31 (0.28)	4

3.3 Multi-panel figure

Figure 3.2 demonstrates two panels within a single numbered figure.

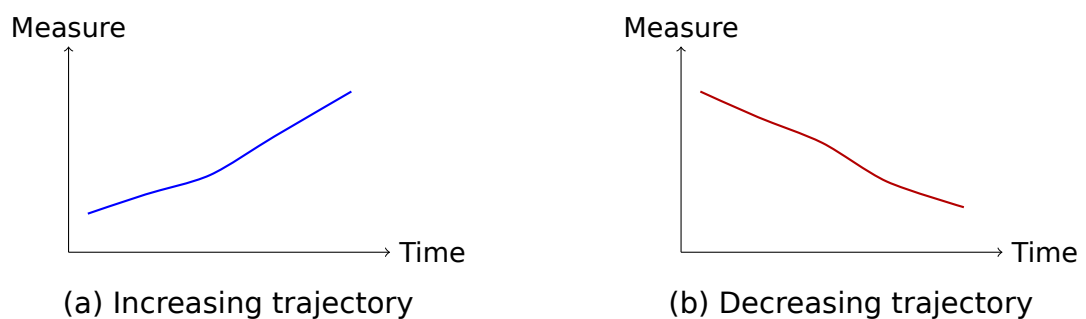


Figure 3.2: Illustrative longitudinal trajectories shown as two panels

3.4 Result summary

The example results chapter now contains a flow diagram, a conventional table and a multi-panel vector figure. Actual analyses should report effect estimates, uncertainty, denominators and missing data alongside clinically meaningful interpretation.

Chapter 4

Discussion

4.1 Principal findings

This example shows how the discussion can refer back to Figure 1.1, Table 2.1 and Equation 1.1 without manually entering their numbers.

4.2 Strengths and limitations

Interpret the findings, limitations and contribution to knowledge here. Possible strengths include prospective data collection, prespecified outcomes, standardised imaging and assessment of repeatability. Possible limitations include selection bias, measurement error, loss to follow-up and limited generalisability beyond the recruited population.

An endnote is useful for relevant background information that is not essential to following the immediate argument.¹

4.3 Implications

The main implication of the hypothetical findings would depend on whether the imaging measure improved prediction sufficiently to justify its cost and complexity. Statistical association alone would not establish clinical utility. Future studies might evaluate calibration, discrimination, decision consequences and acceptability to patients.

4.3.1 Implications for research

Replication in an independent sample would test transportability. Longer follow-up could establish whether an early imaging difference is transient or marks a sustained disease trajectory. Studies should also compare the imaging measure with simpler and less costly alternatives.

4.3.2 Implications for practice

Before clinical adoption, a biomarker requires a defined place in a care pathway. Its benefit would depend on whether the result changes management and whether that change improves an outcome that matters to patients without creating disproportionate cost or burden.

4.4 Chapter summary

The discussion should return explicitly to the research questions, distinguish findings from interpretation, acknowledge alternative explanations and identify the precise contribution to knowledge.

Chapter 5

Conclusion

The completed demonstration confirms that the modular files are combined into one consecutively numbered thesis. Summarise the principal conclusions and implications for research or practice here.

The template separates content from presentation: chapter files contain the scholarly material, `main.tex` controls order and metadata, and `brightonthesis.sty` contains the University formatting rules. This separation allows future regulatory changes to be applied across the complete thesis without editing hundreds of pages individually.

Chapter 6

Formatting Showcase

This chapter exists solely to make the template's features easy to inspect. It can be deleted when beginning the real thesis. Every example is formatted by `brightonthesis.sty`; the chapter file contains content and semantic LaTeX commands, but no local margin, font or page-style settings.

The running header is visible above. It uses smaller type and shows the candidate name, submission year and page number. It appears only in the numbered main text. The formal page number is also retained at the bottom centre. The option that enables the header is set near the top of `main.tex`.

6.1 Heading hierarchy

Use headings to describe the structure of the argument. LaTeX calculates their numbers, adds the required levels to the table of contents and updates cross-references after compilation.

6.1.1 A numbered subsection

This heading demonstrates the second level beneath a chapter. In a long thesis, subsections help readers identify the distinct components of a method, result or argument without making every short paragraph a separate section.

6.1.1.1 A numbered subsubsection

This is the third displayed level beneath a chapter. The stylesheet gives it a smaller, left-aligned heading while retaining automatic numbering. Use this level sparingly so the hierarchy remains easy to follow.

6.1.1.2 A second subsubsection

The next heading number is generated automatically. If a section is moved to another chapter, all its heading, figure, table and equation references update after recompilation.

6.1.2 Paragraphs and inline text

Ordinary paragraphs are typed without formatting commands. A completely blank line in the source begins a new paragraph. The stylesheet applies the approved 12-point font, one-and-a-half spacing, flush-left alignment and additional space between paragraphs consistently across the thesis.

Use emphasis for a term that needs modest emphasis and **bold text** only where a strong visual distinction is genuinely useful. File names and short pieces of code can be shown as `chapters/06-formatting-showcase.tex`. A non-breaking space in a reference, such as Figure 6.1, prevents the label and number from being separated at a line ending.

6.2 Lists

Lists are useful when the items are genuinely parallel. They should not replace connected explanatory prose simply to make the page look more varied.

6.2.1 Bullet points

The `itemize` environment creates bullet points:

- define the population and sampling frame;
- describe acquisition and quality-control procedures;
- identify the primary and secondary outcomes;
- prespecify the statistical analysis; and
- report missing data and uncertainty.

Text following the list returns automatically to the ordinary paragraph style. Punctuation is included here because the bullets form one continuing sentence.

6.2.2 Numbered steps

The `enumerate` environment is appropriate when order matters:

1. Screen potential participants against the eligibility criteria.
2. Obtain informed consent before research procedures begin.
3. Acquire clinical and imaging measurements using the protocol.
4. Perform quality control without reference to the later outcome.
5. Conduct the prespecified analysis and report all relevant results.

6.2.3 Labelled descriptions

A description list pairs a short label with explanatory text:

Population

Adults with symptoms but without established radiographic osteoarthritis at baseline.

Exposure

A quantitative MRI measure obtained using a standardised acquisition and processing pipeline.

Outcome

Change in pain, function or structural status during follow-up.

6.3 Figures and captions

Figures are numbered by chapter, listed automatically in the list of figures and referred to using labels. Captions appear below figures in accordance with the Brighton guidance.

6.3.1 Example figure one: process diagram

Figure 6.1 is a vector process diagram. Because it is drawn in LaTeX, it remains sharp when enlarged in the PDF.

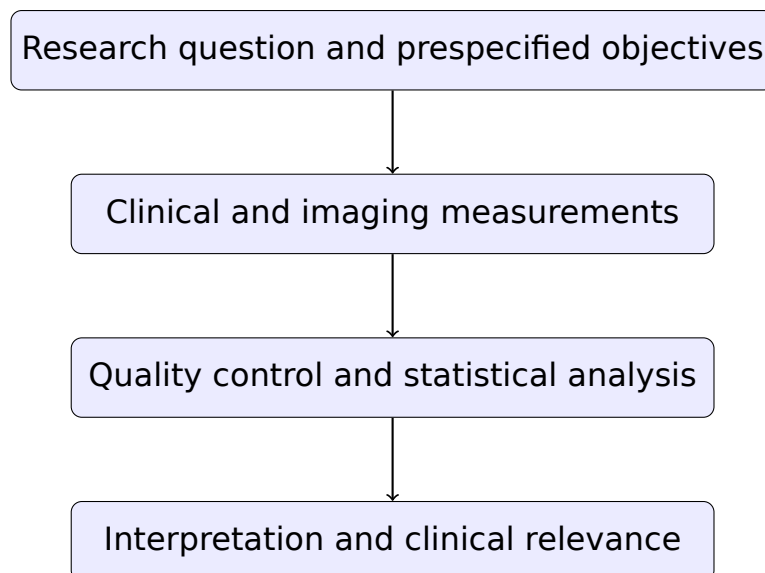


Figure 6.1: Example pathway from research question to interpretation

6.3.2 Example figure two: grouped observations

Figure 6.2 demonstrates a second, visually distinct figure and an automatically generated cross-reference.

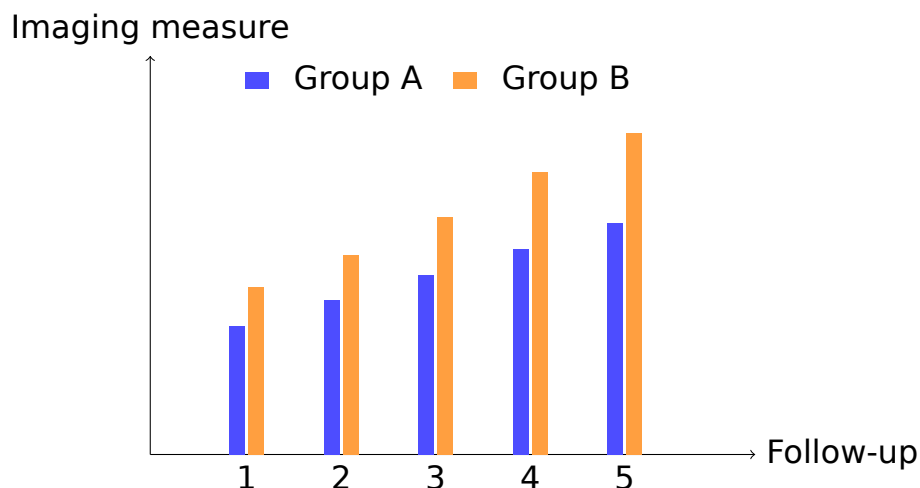


Figure 6.2: Illustrative grouped measurements over five follow-up occasions

When using an external image instead, place the image in a `figures/` folder and replace the drawing code with `\includegraphics`, as shown in the README.

6.4 Table, equation and quotation

6.4.1 Example table

Table 6.1 has its number and title above the table. The row and column headings include enough information for the example to be understood independently of the surrounding paragraph.

Table 6.1: Illustrative measurements and reporting formats

Domain	Example measure	Reporting format
Symptoms	Pain score	Mean (SD)
Structure	Cartilage thickness	Millimetres
Composition	T2 relaxation time	Milliseconds
Function	Walking speed	Metres per second

6.4.2 Example equation

Numbered equations can be referenced in the same way as figures and tables:

$$\Delta Y = Y_{\text{follow-up}} - Y_{\text{baseline}}. \quad (6.1)$$

Equation 6.1 defines a simple change score. Short mathematical expressions such as $n = 120$ can also appear within a sentence.

6.4.3 Example quotation

This indented passage demonstrates the single-spaced quotation environment. Replace it with the quoted material, include a page number where appropriate and provide the corresponding citation.

6.5 Footnote and endnote examples

A footnote marker appears here.¹ Footnotes should contain information needed to understand the nearby text without interrupting the main sentence.

An endnote marker appears here.² The superscript marker beside the preceding sentence points to the numbered entry printed in the separate Endnotes section. It is not printed at the bottom of this page. Endnotes are useful for additional material that can be read separately from the immediate argument.

¹This is a visible example footnote. It is numbered automatically and printed at the bottom of the same page in smaller, single-spaced text.

6.6 Cross-references and chapter summary

This chapter referred to Section 6.1, Figure 6.1, Figure 6.2, Table 6.1 and Equation 6.1 without typing their final numbers. The labels remain stable even if the content moves.

The source demonstrates running headings, multiple heading levels, ordinary paragraphs, inline formatting, bullet points, numbered steps, description lists, two figures, a table, an equation, a quotation, a footnote, an endnote and cross-references. Delete the line `\include{chapters/06-formatting-showcase}` from `main.tex` when this demonstration is no longer required.

Endnotes

¹This is an example endnote. Unlike a footnote, it is collected in the dedicated Endnotes section near the end of the thesis. Endnotes are numbered automatically in the order in which they are called.

²This second example endnote was called from the formatting-showcase chapter. It is collected with the earlier endnote in the dedicated Endnotes section near the end of the thesis.

Glossary

Articular cartilage

The specialised connective tissue covering the ends of bones in synovial joints.

Quantitative imaging

Imaging methods that produce numerical measures of tissue structure or composition.

List of references

Bibliography

- [1] Example Author (2026). Early imaging and joint health. *Journal of Demonstration Studies*, 1(1), 1-10.

Bibliography

Include works consulted but not cited if your discipline requires a separate bibliography. Otherwise delete this file and its corresponding `\include` line in `main.tex`.

Chapter A

Supplementary material

A.1 Example appendix section

Appendix chapters are lettered automatically. Add supplementary methods, tables, questionnaires or other material here.